Task 2: The Greeks

Objective: To build a calculator that outputs these real world greeks $\Delta(\boldsymbol{delta}), \boldsymbol{\Gamma}(\mathbf{Gamma}), \boldsymbol{\Theta}(\mathbf{Theta})$ **and** $\boldsymbol{\nu}(\mathbf{Vega})$ for an ATM (At-the-Money) call and put options using live or historical data.

Data Collection and Methodology:

Underlying Price and Option Chains:-

Collected from the yfinance library in Python to fetch current market data. Ticker data is downloaded using “AAPL” string (Apple) and latest expiry date is used for the option-chains. If the present date is actually the expiry date, then next closest expiry date is used. This is done to keep the Time of maturity positive. Then strike prices(K) for call and put options are obtained for those expiration date.
Risk free rate (r): Current 10-year Treasury Yield is used to set the risk free rate. Treasury Yield is obtained by ticker using string “^TNX”. This ticker automatically gives the last 10-year treasury yield multiplied by 10, which is then divided by 100. The obtained value is then set as risk free rate.

Volatility($\boldsymbol{\sigma}$): Historical volatility was estimated using daily logarithmic returns over the previous 252 trading days.

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Then sigma was calculated as $\sigma = StdDev(r_t)\sqrt{252}$

Spot Price used was the close value of the last minute, using history option in the ticker and period used was last ‘1 min’. This is then used to get the ATM option from call options and strike price is obtained.

Greeks are calculated by using the formulae

Delta (Δ):

- *Call Formula:* $\Delta_c = N(d_1)$
- *Put Formula:* $\Delta_p = N(d_1) - 1$

Gamma: $\Gamma = \frac{N'(d_1)}{S\sigma\sqrt{T}}$

Theta (Θ):

- *Call Formula:* $\Theta_c = -\frac{SN'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$
- *Put Formula:* $\Theta_p = -\frac{SN'(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}N(-d_2)$

Vega: $\nu = S\sqrt{T}N'(d_1)$

$$d_1 = \frac{\ln\ \frac{S}{K}) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Scipy.stats.norm is used for standard normal distribution functions and a pandas DataFrame table is printed for these calculated values. At the time of writing, the table is as follows

	Delta	Gamma	Theta	Vega
Option Type				
CAll	0.109538	0.135035	-303.856278	29.310407
PUT	-0.890462	0.135035	-300.961859	29.310407